

# GREENLIFT: Components and Connection Detailing

## 1. Components List

| Subsystem / Function | Component Specification                           |
|----------------------|---------------------------------------------------|
| Structural Chassis   | T-Slot Aluminum Extrusions                        |
| Rotating Cross-Hub   | Grey PVC pipes & 90-degree elbows                 |
| Plant Suspension     | Free-spinning pivot pins & hanging baskets        |
| Smart Base Brain     | Arduino Uno R3                                    |
| Drive System         | High-Torque DC Geared Motor                       |
| Watering Payload     | 12V Micro Submersible Pump & 1-Channel Relay      |
| Water Source         | Rainwater Collection Reservoir & Blue Tubing      |
| Moisture Payload     | Analog Capacitive Soil Moisture Probe             |
| User Controls        | Push-Button (Watering) & Manual Switch (Rotation) |
| Alarm Suite          | Digital Buzzer Module                             |
| Power Management     | Solar Panel Array (Concept) & Battery Packs       |

## 2. Connection Detailing / Pinout Mapping

### System A: Direct-Drive Rotation Circuit

| Connection / Pin | Component / Function                                           |
|------------------|----------------------------------------------------------------|
| Power Circuit    | Wired directly to Solar Power Source / Battery                 |
| Manual Switch    | Independent ON/OFF switch to manually rotate the DC gear motor |

### System B: Arduino Uno (Smart Base Station Controller)

| Pin / Connection | Component / Function                                     |
|------------------|----------------------------------------------------------|
| Digital Pin 7    | 1-Channel Relay Module (Triggers the 12V Rainwater Pump) |

| Pin / Connection     | Component / Function                                     |
|----------------------|----------------------------------------------------------|
| <b>Digital Pin 4</b> | Digital Buzzer Module (Interval Alarm Monitor)           |
| <b>Digital Pin 2</b> | Push-Button Input (Manual Rainwater Dispensing Trigger)  |
| <b>Analog Pin</b>    | Capacitive Soil Moisture Sensor (Checks plant hydration) |
| <b>Power</b>         | 5V and GND connected to modules                          |